

SESSION 9

Email Sending in Django

Topics Covered: SMTP setup, Django email functions, HTML email templates

Activities: Send welcome email on signup

Task 1: Configure SMTP email settings in settings.py to use Gmail for sending emails and test using send_mail function.

Explanation & Approach: *Django leverages standard SMTP protocols with App Passwords for secure email delivery.*

```
# settings.py
EMAIL_BACKEND = 'django.core.mail.backends.smtp.EmailBackend'
EMAIL_HOST = 'smtp.gmail.com'
EMAIL_PORT = 587
EMAIL_USE_TLS = True
EMAIL_HOST_USER = 'your_email@gmail.com'
EMAIL_HOST_PASSWORD = 'your_app_password' # Generated from Google Account

# Shell Test Script (python manage.py shell)
from django.core.mail import send_mail
send_mail(
    'Test Email',
    'This is a test email sent from Django SMTP setup.',
    'your_email@gmail.com',
    ['recipient@example.com'],
    fail_silently=False,
)
```

Task 2: Create Django view function send_password_reset_email(request, user_email) that sends password reset link using send_mail.

Explanation & Approach: *Generates dynamic reset links for password recovery workflows.*

```
# views.py
from django.core.mail import send_mail
from django.shortcuts import render, HttpResponseRedirect

def send_password_reset_email(request, user_email):
    reset_url = f"http://127.0.0.1:8000/accounts/reset/token12345/"
    subject = "Password Reset Request - Paytm Clone"
    message = f"Hello,\n\nYou requested a password reset. Click link below to\nreset:\n{reset_url}\n\nIf you did not request this, please ignore."

    send_mail(subject, message, 'noreply@paytm.com', [user_email])
    return HttpResponseRedirect("Password reset email sent successfully!")
```

Task 3: Design HTML email template order_confirmation.html for Swiggy-style order confirmation and send using EmailMultiAlternatives.

Explanation & Approach: *EmailMultiAlternatives* sends multi-part MIME messages containing both plain text fallback and rich HTML layouts.

```
# views.py
from django.core.mail import EmailMultiAlternatives
from django.template.loader import render_to_string

def send_order_confirmation(user_email, order_data):
    subject = "Swiggy - Order Confirmation"
    text_content = f"Thank you for your order #{order_data['order_id']}!"

    # Render HTML template to string
    html_content = render_to_string('emails/order_confirmation.html', {'order':
order_data})

    msg = EmailMultiAlternatives(subject, text_content, 'orders@swiggy.com', [user_email])
    msg.attach_alternative(html_content, "text/html")
    msg.send()
```

Task 4: Modify HTML email template to include user name and order details dynamically using Django template variables.

Explanation & Approach: *Context variables injected into render_to_string* allow personalized transactional emails.

```
<!-- templates/emails/order_confirmation.html -->
<!DOCTYPE html>
<html>
<head>
    <style>
        body { font-family: Arial, sans-serif; background-color: #f4f4f4; padding: 20px; }
        .card { background: white; padding: 20px; border-radius: 8px; }
        .orange { color: #fc8019; }
    </style>
</head>
<body>
    <div class="card">
        <h2 class="orange">Swiggy Order Confirmed!</h2>
        <p>Hi <strong>{{ order.customer_name }}</strong>,</p>
        <p>Your order from <strong>{{ order.restaurant_name }}</strong> has been
placed!</p>
        <hr>
        <h4>Order Summary (#{ order.order_id }):</h4>
        <ul>
            {% for item in order.items %}
                <li>{{ item.name }} x {{ item.qty }} - ₹{{ item.price }}</li>
            {% endfor %}
        </ul>
        <h3>Total Paid: ₹{{ order.total_amount }}</h3>
```

```

    </div>
</body>
</html>

```

Task 5: Use ChatGPT or Copilot to generate catchy subject line and body for 'Welcome to IPL Fantasy League' HTML email and implement sending logic.

Explanation & Approach: *Integrating AI copy into structured EmailMultiAlternatives sends polished engagement emails.*

```

# views.py
from django.core.mail import EmailMultiAlternatives

def send_ipl_welcome_email(user_email, username):
    subject = "🔥 Get Ready for the Ultimate Showdown! Welcome to IPL Fantasy League 2026 🔥"
    text_content = f"Hey {username}, welcome to IPL Fantasy League! Pick your XI and win grand prizes."

    html_content = f"""
    <html>
        <body style="background-color: #0b132b; color: white; padding: 30px; font-family: sans-serif;">
            <div style="max-width: 500px; margin: auto; border: 2px solid #5bc0be; padding: 20px; border-radius: 12px; text-align: center;">
                <h1 style="color: #6fffe9;">🔥 Welcome to IPL Fantasy!</h1>
                <p>Hey <strong>{username}</strong></p>
                <p>The stadium is roaring and your dream team awaits! Build your playing XI now and climb the global leaderboard.</p>
                <a href="http://127.0.0.1:8000/fantasy/create-team/" style="background: #5bc0be; color: #0b132b; padding: 12px 24px; text-decoration: none; font-weight: bold; border-radius: 6px; display: inline-block; margin-top: 15px;">Create Your XI Now</a>
            </div>
        </body>
    </html>
    """

    msg = EmailMultiAlternatives(subject, text_content, 'fantasy@ipl.com', [user_email])
    msg.attach_alternative(html_content, "text/html")
    msg.send()

```